power_input

File: power_input.kicad_sch

boost_charger

File: boost_charger.kicad_sch

cap_bank

File: cap_bank.kicad_sch

pulse_switch

File: pulse_switch.kicad_sch

flyback

File: flyback.kicad_sch

sense_gates

File: sense_gates.kicad_sch

mcu_status

File: mcu_status.kicad_sch

OmniMarble

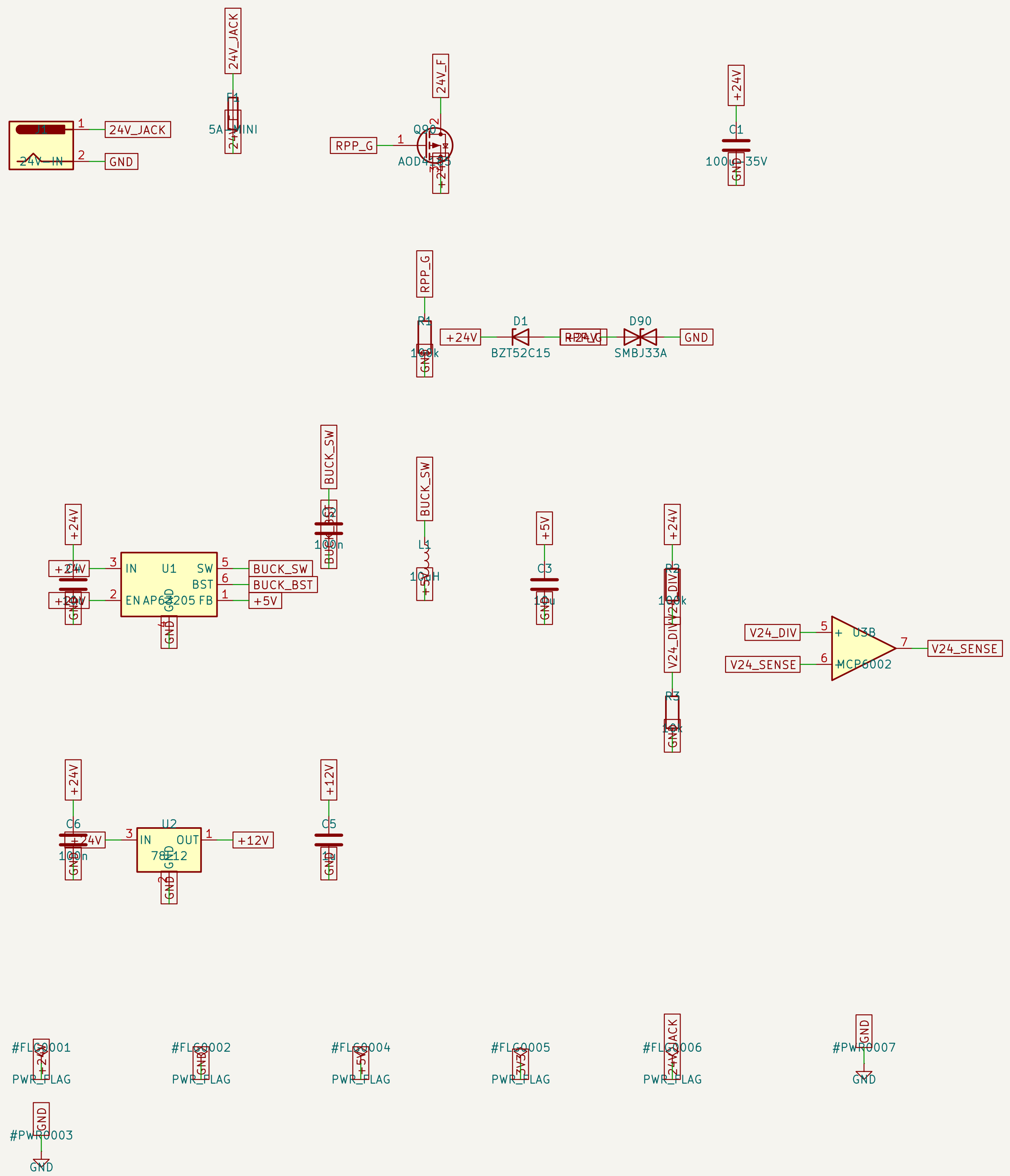
Sheet: /

File: omnimarble-driver.kicad_sch

Title: OmniMarble 60V SELV coilgun driver

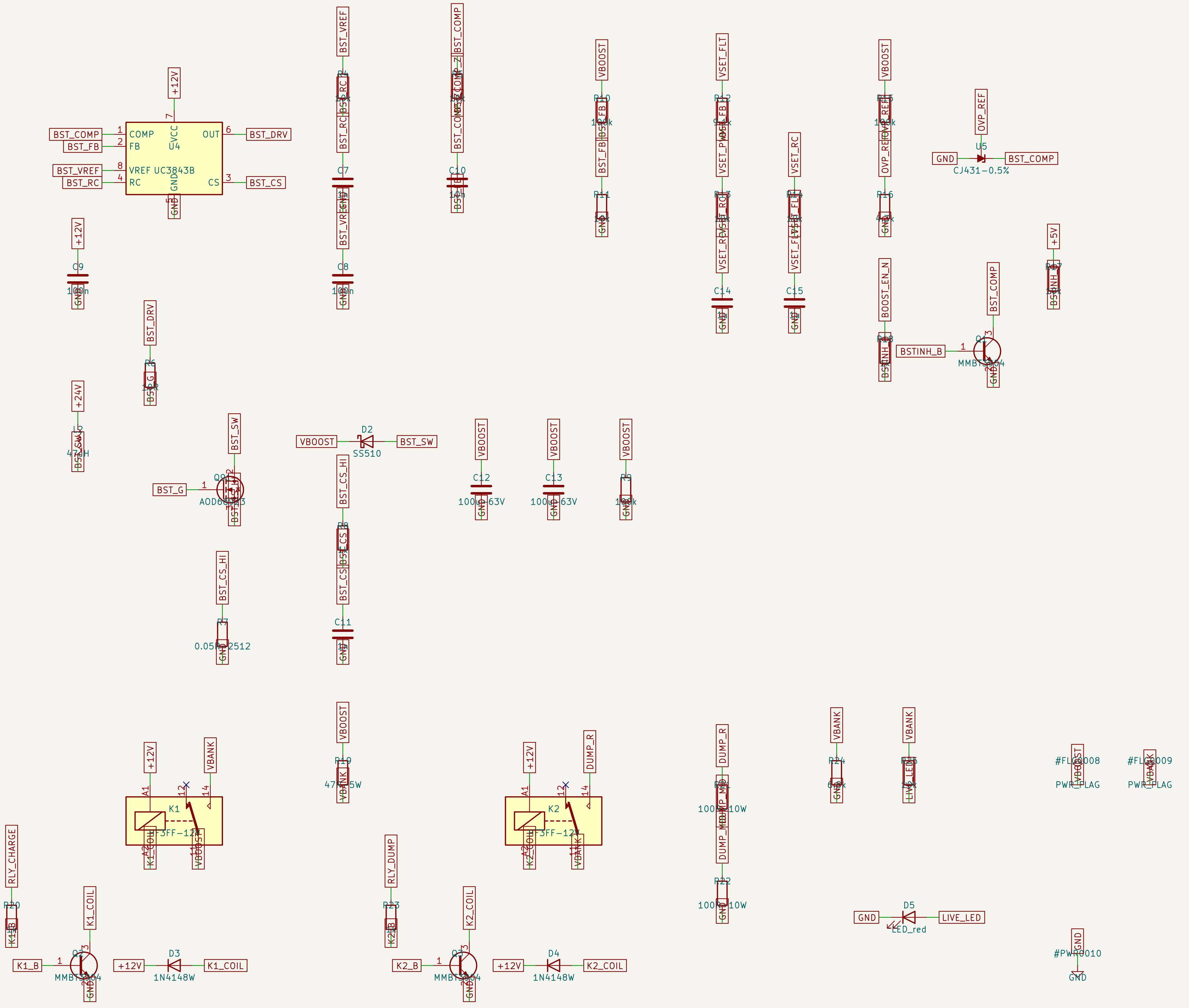
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/8

out, protection, 5V buck, 12V rail



OmniMarble
Sheet: /power_input/
File: power_input.kicad_sch
Title: omnimarble-driver / power_input
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 2/8

5V operating (63V–rated bank), OVP, relays, bleed

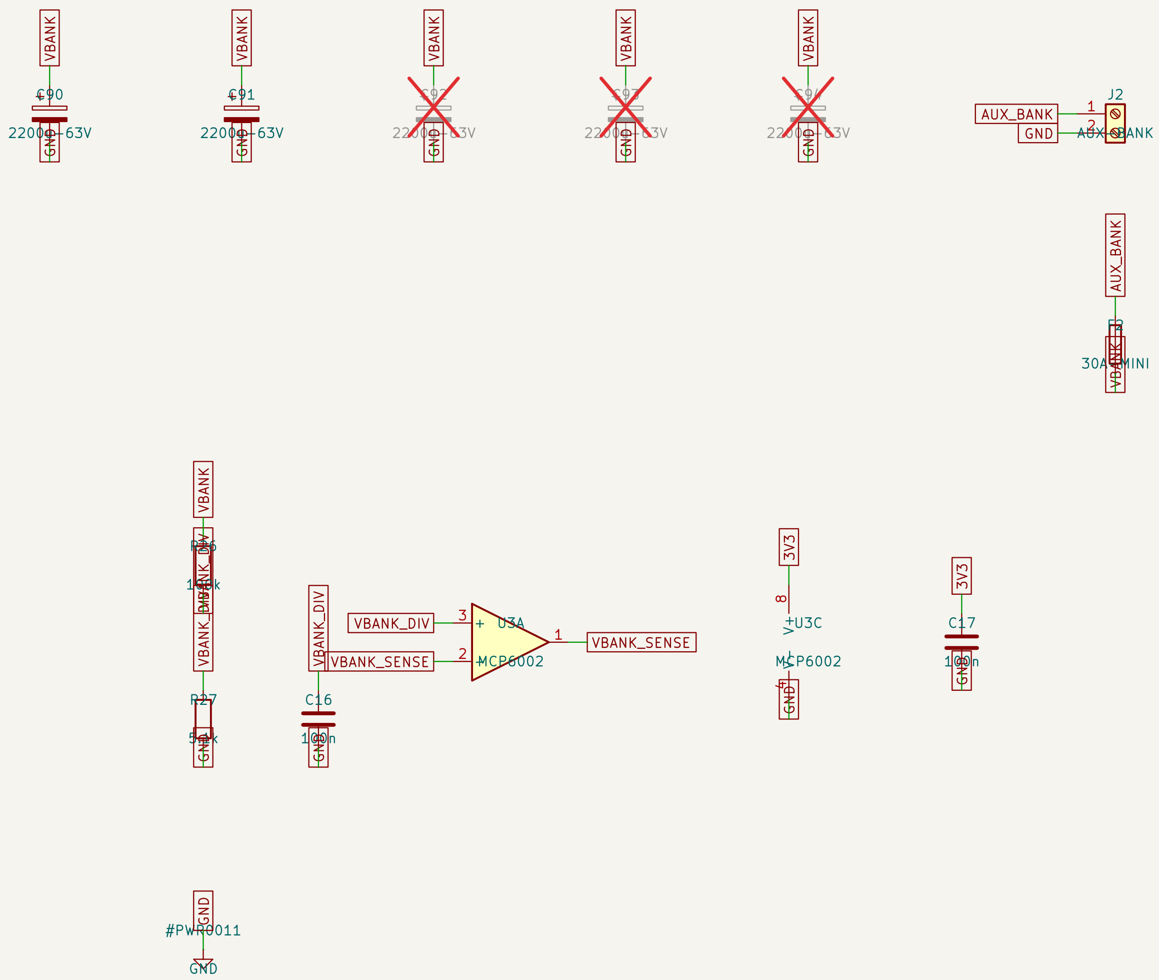


OmniMarble
Sheet: /boost_charger/
File: boost_charger.kicad_sch

Title: omnimarble–driver / boost_charger

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 3/8

2200uF/63V snap-in (populate per bank table)

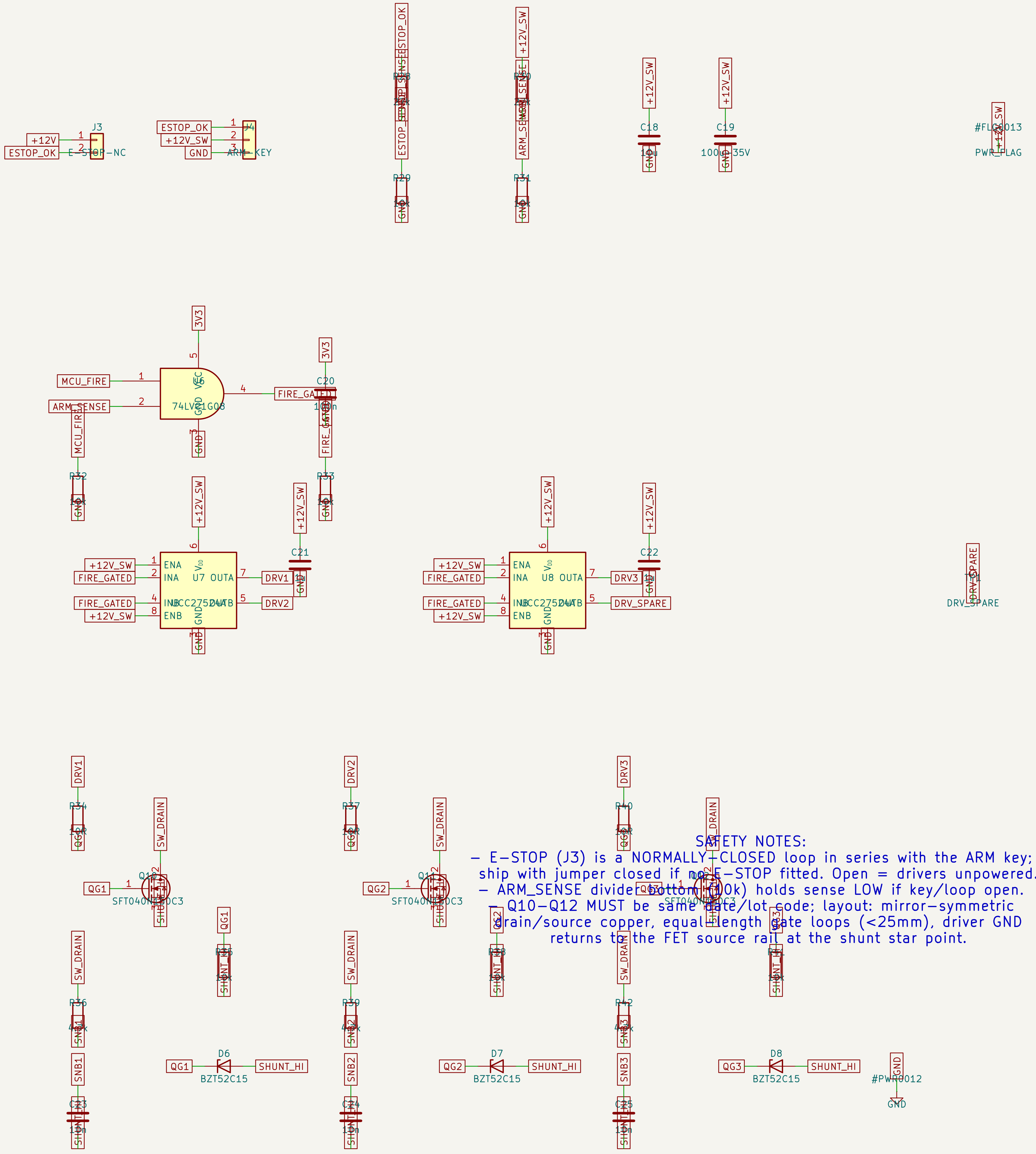


OmniMarble
Sheet: /cap_bank/
File: cap_bank.kicad_sch

Title: omnimarble-driver / cap_bank

Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 4/8

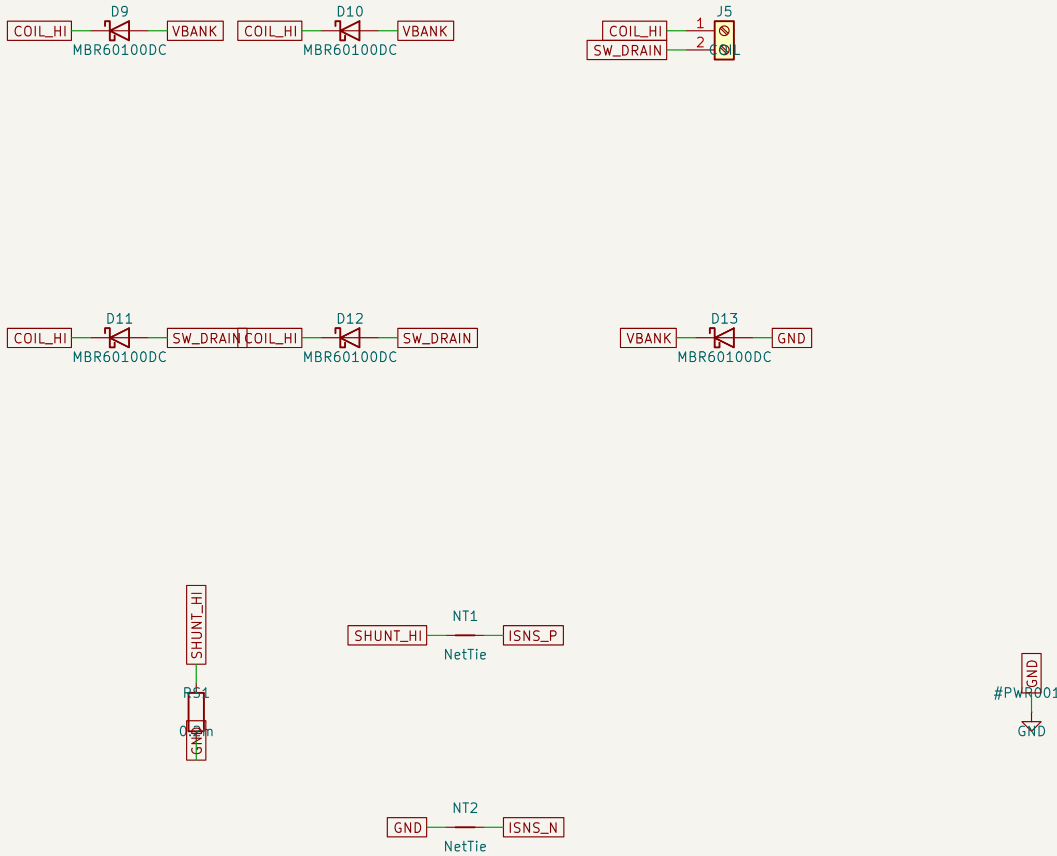
T040N150C3, UCC27524A drive, ARM interlock



SAFETY NOTES:
- E-STOP (J3) is a NORMALLY-CLOSED loop in series with the ARM key;
ship with jumper closed if no E-STOP fitted. Open = drivers unpowered.
- ARM_SENSE divider bottom (10k) holds sense LOW if key/loop open.
- Q10-Q12 MUST be same date/lot code; layout: mirror-symmetric
drain/source copper, equal-length gate loops (<25mm), driver GND
returns to the FET source rail at the shunt star point.

OmniMarble
Sheet: /pulse_switch/
File: pulse_switch.kicad_sch
Title: omnimarble-driver / pulse_switch
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 5/8

blocking -> COIL -> switch -> shunt -> GND

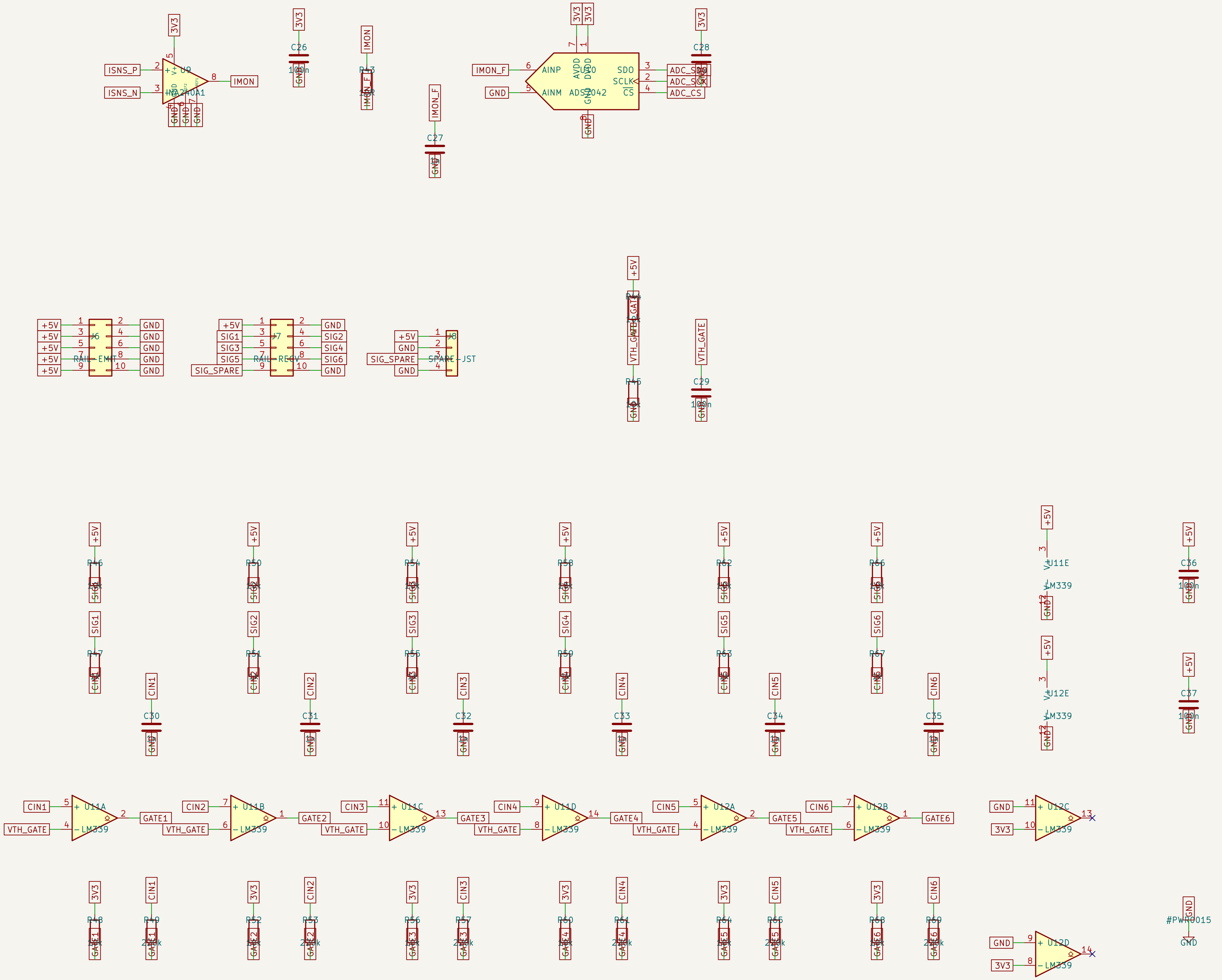


– validated envelope, swept in calcs.py at 11000uF / 55V:
>= 1 uH AND R_total >= 90 mOhm
oil DC resistance + leads/lugs; keep leads short)
fault: 30T demo coil, 12.406 uH, 110 mOhm total
se can exceed the 600A switch design point – DO NOT FIRE.

OmniMarble
Sheet: /flyback/
File: flyback.kicad_sch
Title: omnimarble-driver / flyback

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/8

42 current capture; 6x IR gate comparators



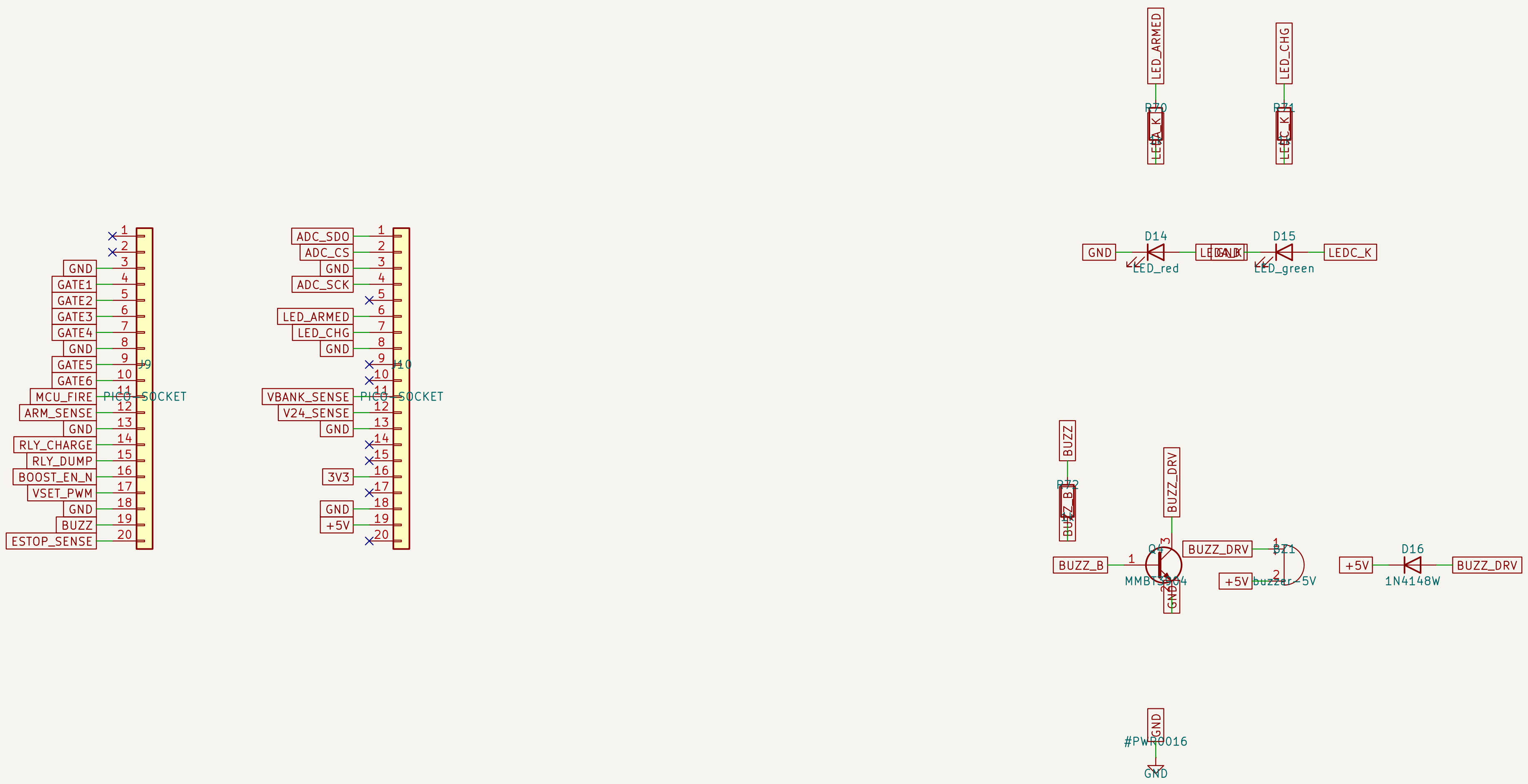
IR GATE TRUTH TABLE:
T -> phototransistor ON -> SIGx ~0V -> comparator OUT LOW
-> phototransistor OFF -> SIGx ~5V -> comparator OUT HIGH
GH = marble in beam. Firmware timestamps the RISING edge.
pen-collector outputs pulled to 3V3: Pico-safe levels)

OmniMarble
Sheet: /sense_gates/
File: sense_gates.kicad_sch

Title: omnimarble-driver / sense_gates

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 7/8

0), MCU–agnostic breakout, relay/status drive



OmniMarble
Sheet: /mcu_status/
File: mcu_status.kicad_sch
Title: omnimarble-driver / mcu_status
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 8/8